# Computational methods for automated analysis of Malaria using Blood Smear Images

**A Major Project for the partial fulfillment of the requirement for the award of the degree of**

## MASTER OF COMPUTER APPLICATIONS

SUBMITTED

BY

**Nandini.K**

**Roll No.: 21691F0096**

Under the Guidance of

**Dr. S.JANSI M.C.A., Ph.D.,**

Assistant Professor

Department of Computer Applications




**DEPARTMENT OF COMPUTER APPLICATIONS**

MADANAPALLE INSTITUTE OF TECHNOLOGY &SCIENCE

UGC-AUTONOMOUS

(Approved by AICTE, NewDelhi & Affiliated to JNTUA, Ananthapuramu)

www.mits.ac.n

2022-2023

**MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE**

**(UGC-AUTONOMOUS)**

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)

**DEPARTMENT OF COMPUTER APPLICATIONS**

## BONAFIDE CERTIFICATE

This is to certify that the project work entitled "Computational methods for automated analysis of malaria using Blood Smear Images" is a bonafide work carried out by NANDINI.K, Roll No:21691F0096, submitted in the partial fullfillment of the requirements for the award of the degree of Master of Computer Applications in Madanapalle Institute of Technology &Science, Madanapalle, affiliated to Jawaharlal Nehru Technological University Anantapur, Anantapuramu during the academic year 2022-2023.

| PROJECT GUIDE | HEAD OF THE DEPARTMENT |
|---|---|
| **Dr. S. Jansi, M.C.A., Ph.D.,** | **Dr. N. Naveen Kumar, M.C.A., Ph.D.,** |
| **Assistant Professor** | **Associate Professor** |
| **Department of Computer Applications** | **Department of Computer Applications** |

**Submitted for viva voice examination held on_____**

| Internal Examiner | External Examiner |
|---|---|
| Date: | Date: |

## DECLARATION

I, NANDINI.K(Roll No.:21691F0096) hereby declare that the project entitled "Computational methods for automated analysis of malaria using Blood smear Images" is done by me under the guidance of Dr.S.Jansi, M.C.A., Ph.D., submitted in partial fullfillment of the requirements for the award of degree of Master of Computer Applications at MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE Madanapalle, Affiliated to Jawaharlal Nehru Technological University Anantapur,Anantapuramu during the academic year 2022-2023.This work has not been submitted by anybody towards the award of any degree.

Date:

Place: Madanapalle

Signature of the Student

(Nandini.K)
(Roll No.: 21691F0096)

## ACKNOWLEDMENT

First, I must thank the almighty who has granted me knowledge, wisdom, strength and courage to serve in this world and to carry out my project work in a successfully way**.**

Next, I have to thank my parents who encouraged me to undergo this MCA course and do this project in a proper way.

I express my sincere thanks to **Dr.N.Vijaya Bhaskar Chowdary, Ph.D.,** Secretary& Correspondent, Madanpalle Institute of Technology & Science for his continuous encouragement to wards practical education and constant support in all aspects which includes the provision of very good infrastructure facilities in the institute.

It is my duty to thank our Principal, **Dr. C. Yuvaraj, M.E, Ph.D.,**Madanapalle Institute of technology & science, for his guidance and support at the time of my course and project.

I also wish to express my thanks to our vice principal, **Dr.P.Ramanathan, Ph.D.,** for his continuous support though out our MCA career.

It is my foremost duty to thank Head of the Department **Dr. N. Naveen Kumar, M.C.A., Ph.D.,**who gave me constant support during the project time and continuous encouragement to words the completion of the project successfully.

I thank my faculty guide **Dr.S.JANSI, M.C.A., Ph.D.,** for her continuous support by conducting periodical reviews and guidance until the completion of the project in as successful manner.

**NANDINI.K**

# ABSTRACT

Malaria is a disease which, despite being present for over a century, still claims a significant number of lives every year. It is the primary reason for most of the causalities across the world. It is presently rated as a significant cause of the high death rate worldwide compared with other diseases. The main objective of this project is to reduce the death rate by predicting malaria at an early stage. The Existing system, AlexNet which is very expensive and give less accuracy in classifying malaria infected cells. In Proposed system, the different computer-assisted techniques such as image acquisition, image pre-processing, image segmentation, feature extraction and classification for the detection of malaria using blood smear images. These image processing and computational intelligence techniques are used to identify whether the red blood cells are infected or not infected. With the help of evaluation metrics such as accuracy and sensitivity the performance of image processing algorithms has been analysed for classification of malaria infected and non-infected cells.



**Keywords**: Malaria, Blood Smear images, Convolutional Neural Network, Feature Extraction, Classification.

# TABLE OFCONTENTS


| CHAPTER | TITLE | PAGE NO. |
|---|---|---|
| I | **INTRODUCTION** | |
| 1.1 | About Project | 7 |
| II | **SYSTEM ANALYSIS** | |
| 2.1 | Existing System | 8 |
| | Disadvantages of Existing System | 8 |
| 2.2 | Proposed System | 8 |
| | Advantages of Proposed System | 8 |
| 2.3 | Hardware and Software Specification | 9 |
| 2.4 | Feasibility Study | 9 |
| III | **SYSTEM DESIGN** | |
| 3.1 | Modules and Description | 10-13 |
| 3.2 | UML Diagrams | 14 |
| | Use case Diagram | 14 |
| | Class Diagram | 15 |
| | Component Diagram | 15 |
| | Activity Diagram | 16 |
| IV | **SYSTEM IMPLEMENTATION** | |
| 4.1 | System Implementation | 17-18 |
| 4.2 | Sample Code | 19-21 |
| 4.3 | Screen Shots | 22-25 |
| V | **SYSTEM TESTING** | |
| 5.1 | Testing Description | 26 |
| VI | **CONCLUSION & FUTURE ENHANCEMENTS** | |
| 6.1 | Conclusion | 27 |
| 6.2 | Future Enhancements | 27 |
| | **REFERENCES** | 28-29 |
| | **PUBLICATION** | 30 |

# CHAPTER- I

# INTRODUCTION

## 1.1 About the Project

Malaria is a significant burden on our healthcare system and it is the major cause of death in many developing countries. It is endemic in some parts of the world which means that the disease is regularly found in the region. Therefore, early testing is necessary to detect malaria and save lives. Thus, it gives us the motivation to make malaria diagnosis more effective and faster. A specialized technology proves essential to combat this problem. The detection and diagnosis of malaria parasitic infection is even more necessary to be efficient, reliable and accurate in order for us to tackle the endemic and control or reduce the number of deaths annually. The applications of deep learning which include models such as the CNN has various advantages over the manual testing and diagnosis. Malaria is a life-threatening disease. It causes fever, vomiting, headaches and in severe cases it can cause coma or even death. The disease is commonly transmitted by female anopheles' mosquitoes. So early and accurate diagnosis of malaria is an effective treatment. Convolutional Neural Networks have emerged as powerful tools for malaria detection.

# CHAPTER- II

# SYSTEM ANALYSIS

## 2.1 Existing System:

Transfer learning algorithms based on well-established network architectures such as AlexNet is developed in the research for the classification of the malaria-infected cell images. The study proposed by A. Molina et. al. aims to develop an ML approach able to discriminate malaria-parasitised cells from normal cells and from cells infected by other diseases. ML models including random forest and adaptive boosting are proposed to predict malaria using patient information.

### 2.1.1 Disadvantages of Existing System:

- Existing techniques are computationally expensive.
- Existing techniques are sensitive to variation in illumination.
- Some of the existing systems are computationally fast but accuracy wise a little bit lesser.

## 2.2 Proposed system

CNNs are a type of deep neural networks based on a multilayer perception. Their fundamental architecture includes an input layer, one or more hidden layers, and an output layer. The input layer represents the image being tested. The hidden layer consists of the convolutional and the pooling layer which alternate. The output layer includes a flattening function, a fully connected layer, and a softmax function. The models are developed using thin blood smear images. A classifier that successfully performs the recognition of infected cells can add great value to traditional medical methods of malaria identification and treatment. Furthermore, computer-aided detection and diagnosis in medical imaging offers a beneficial second opinion to the doctors and assists them in the screening process.

### 2.2.1 Advantages of Proposed System:

- **Accuracy:** The proposed system achieves the greater accuracy than the existing system models.
- **Complexity:** The proposed system has low complexity.

- **Cost:** The proposed system is computationally not so expensive than the existing system models.

## 2.3 Hardware and Software Specification:

The hardware and software specification for the development of the system is given below.

**Hardware Specification:**

- System : I3 Processor
- Hard Disk : 500 GB
- RAM : 4GB

**Software Specification:**

- Operating System : Windows10
- Coding language : Python

## 2.4 Feasibility Study:

The feasibility study proposes one or more feasible conceptual solutions to the problems set of the project. It indicates what the inputs are needed by the system and what outputs will be produced.

- Technical Feasibility
- Economical Feasibility

### 2.4.1 Technical feasibility

- It refers to the details of how you intend to service to the people. Technical Feasibility is carried out to determine whether the project is feasible in terms of software, hardware to handle the completion of the project.

### 2.4.2 Economical feasibility

- It refers to the advantages or outcome are deriving the no cost spending for developing project. Assess the potential cost savings that could be achieved through the implementation of a CNN-based malaria detection system. Consider the current costs associated with alternative diagnostic methods, such as manual microscopy, rapid diagnostic tests, or other automated systems.

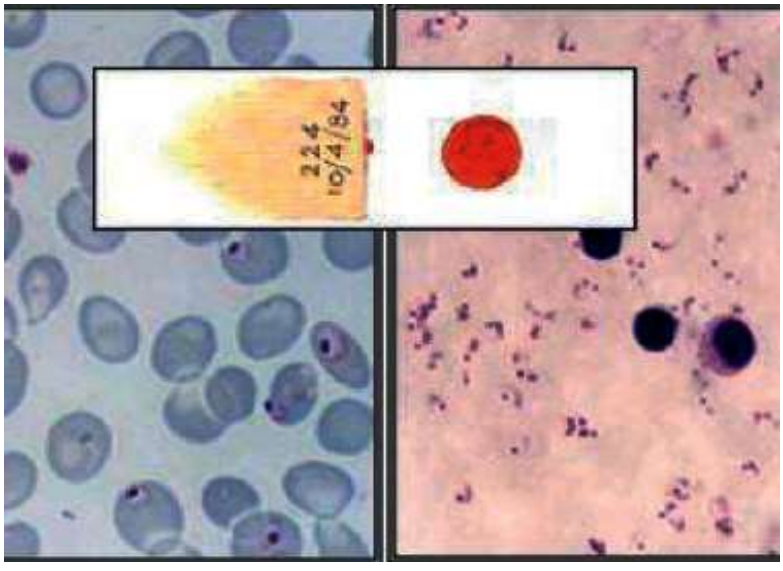
# CHAPTER-III
# SYSTEM DESIGN

## 3.1 Modules and Description:

1. Image Acquisition
2. Pre-Processing
3. Segmentation
4. Feature Extraction
5. Classification

### 3.1.1 Image Acquisition:

The images of blood smear samples are required to detect malaria in a patient using computer vision image processing and machine learning techniques. Each patient's blood smear sample is distributed into two parts: thick and thin blood smear images. Different sources collect digital blood smear images, and this process is called the image acquisition**.**



### 3.1.2 Pre-Processing

- The main aim of this Pre-processing is to clean the cell images. This will remove unwanted noise like redundant image pixels.
- It is mainly useful for improving the quality of an image.
- The Pre-Processing Techniques are
    - ❖ Noise Reduction
    - ❖ Rescaling (Min-Max Normalization)
    - ❖ Standardization

➢ **Noise Reduction:**

Noise reduction is the process of removing the noise from image. For Noise removal the most popular approaches such as mean Filter and Weiner filters are used.

**Mean Filter:** Mean filtering is simply to replace each pixel value in an image with the mean value of its neighbors, including itself. This has the effect of eliminating pixel values which are unrepresentative of their surroundings. It is used to remove noise from blood smear images without affecting the edges.

**Weiner Filter:** It removes the additive noise and inverts the blurring simultaneously. It is used to enhance the quality of blurred images.

➢ **Re-Scaling:**

The patches of images were rescaled to map the range of features to 0 to 1. Since the red blood cell patches are 8-bit RGB color images, the maximum pixel value is 255 and the minimum is 0**.**

➢ **Standardization:**

Another pre-processing technique named feature standardization. Here the values of each feature in the data is rescaled to have mean 0 and standard deviation 1. Earlier, each RBC image had pixel values between 0 to 255. After standardization of RBC patch, images have been rescaled to have mean 0 and standard deviation of 1.

### 3.1.3 Segmentation:

Image Segmentation involves converting an image into a collection of pixels that are represented by a labelled image. The Image is segmented to extract the region of interest from the image. The image segmentation techniques are

- ❖ Cluster based Segmentation
- ❖ Watershed Segmentation

**Cluster based Segmentation**

It divides the image into several components with common pixel characteristics. It is a method of pixel-wise segmentation. There are two approaches for performing the segmentation by clustering.

- Clustering by Merging
- Clustering by Divisive

**Watershed Segmentation**

Watershed segmentation is an image segmentation algorithm that segments an image into several regions. The segmentation process will take the adjacent pixels of the image as an important reference to connect pixels with similar spatial positions.

### 3.1.4 Feature Extraction:

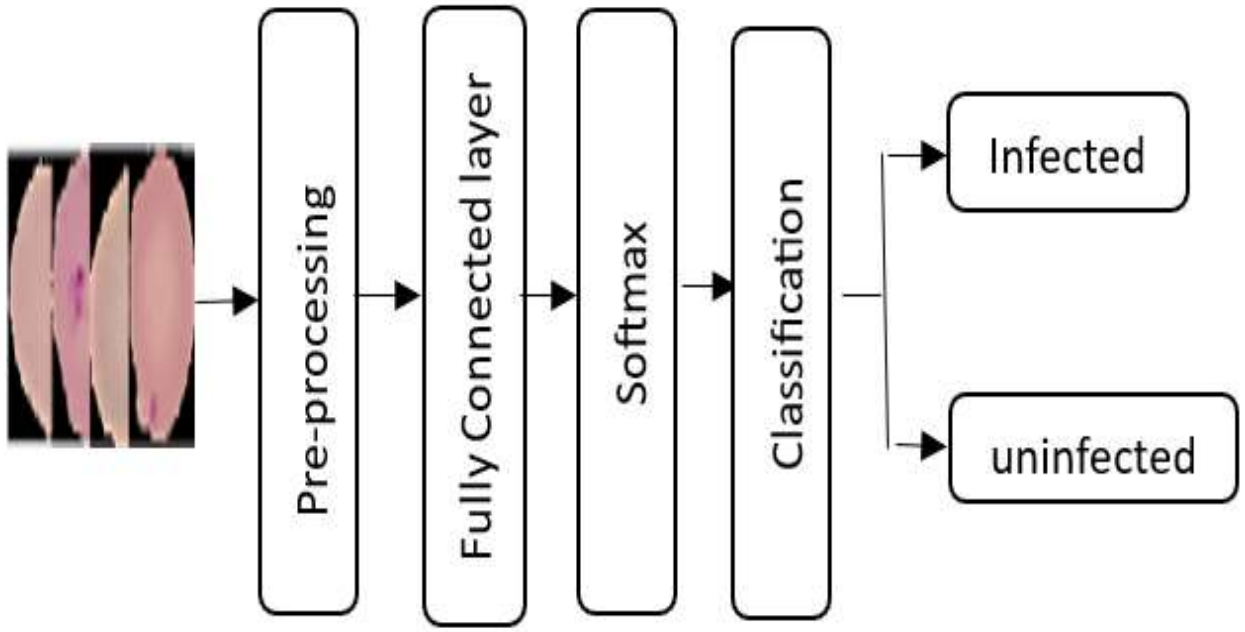
The segmented image is given as a input and we get the feature vectors which are useful for classification task. The objective of feature extraction is to recognize and characterize an object whose dimensions are very nearest or similar for objects in the same class and different for objects from a different class. It reduces the computational complexity of the other processes and provides accurate and reliable recognition to unknown, unrecognized data**.**

- Color Features
- Morphological Features

### 3.1.5: Classification

It is used to classify the features that are extracted from the image into various classes. Convolutional Neural Network is the most popular neural network model being used for image classification. Neural network system consists of several layers of neurons that are inter connected to each other. To activate a neural system input image is given to the input layer and a weight is assigned to each input node. All the nodes are connected each other so input node will activate the other nodes. It will continue until the output node will activate. After train neural network, the system is ready to recognize the cells whether infected or uninfected. The first step here will be to resize all images as CNN cannot train images of different sizes. We compute the mean for both the dimensions and resize all the images.

The architecture of malaria detection using CNN layers is shown below.



## 3.2 UML Diagrams

- The Unified Modelling Language is a standard language for specifying, visualization, Construing and documenting the artefacts of software system, as well as for business modelling and other non-software systems.

1. Use case Diagram
2. Class Diagram
3. Component Diagram
4. Activity Diagram

### 3.2.1 Use case Diagram

A use case diagram is a visual representation of the interactions between actors and a system, showcasing the various use cases or functionalities of the system.

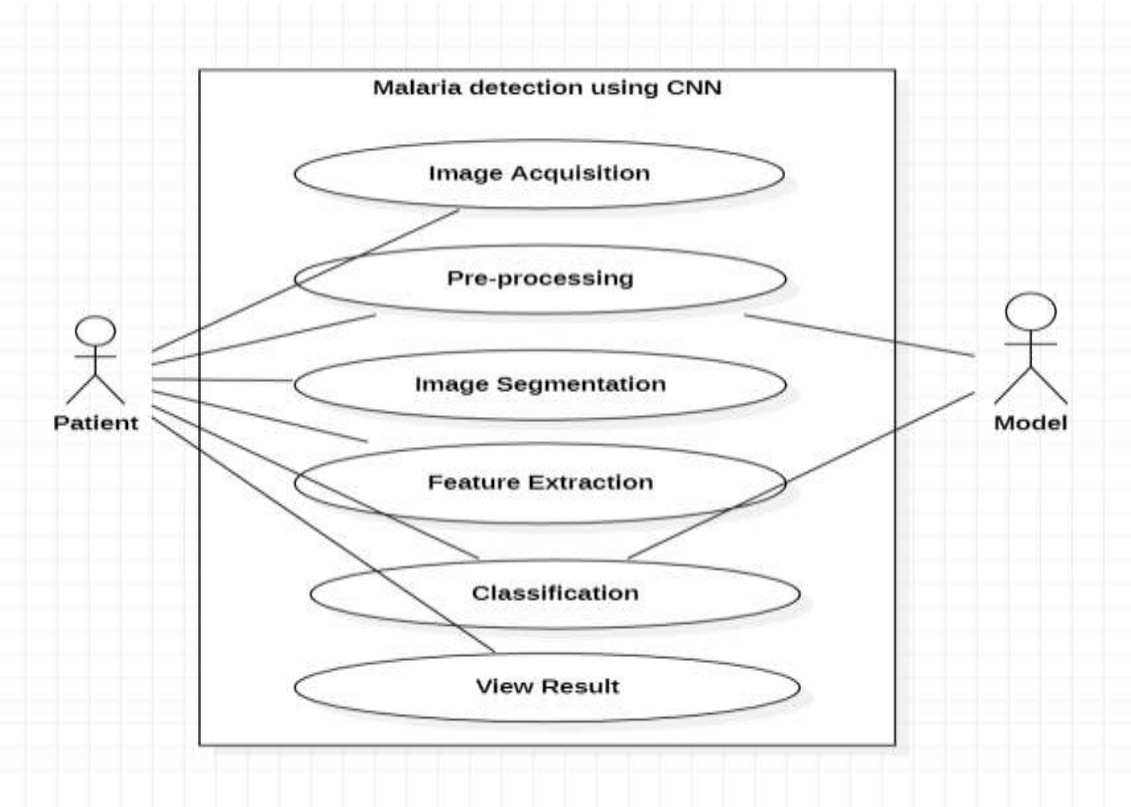



**Fig 3.2.1 Use case Diagram**

### 3.2.2 Class Diagram

Class diagram is type of static structure diagram that describes the structure of a system and showing the system classes, their attributes, operations and the relationship among the classes**.**

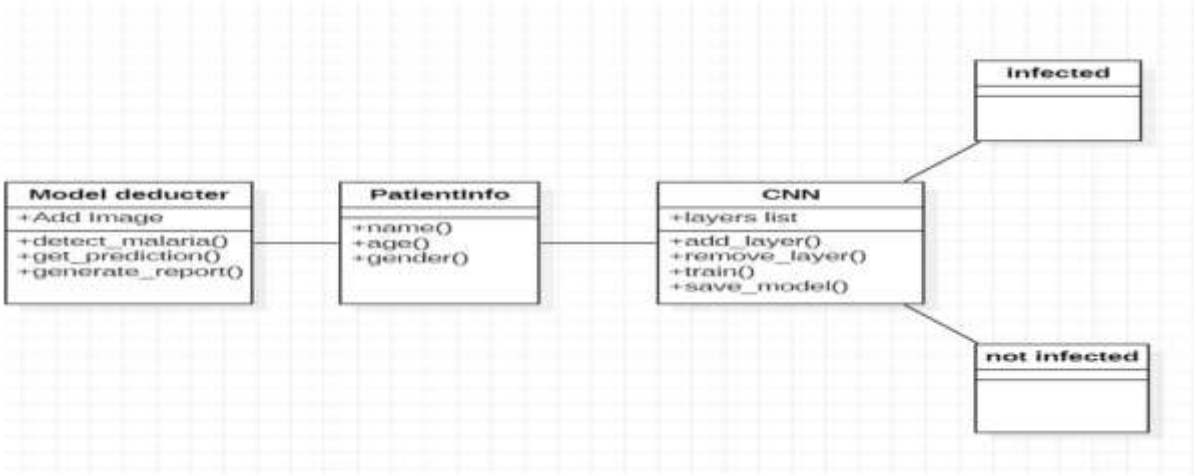



**Fig 3.2.2 Class Diagram**

### 3.2.3 Component Diagram

Component diagrams show the structure of the software system, which describes the software components, their interfaces, and their dependencies.

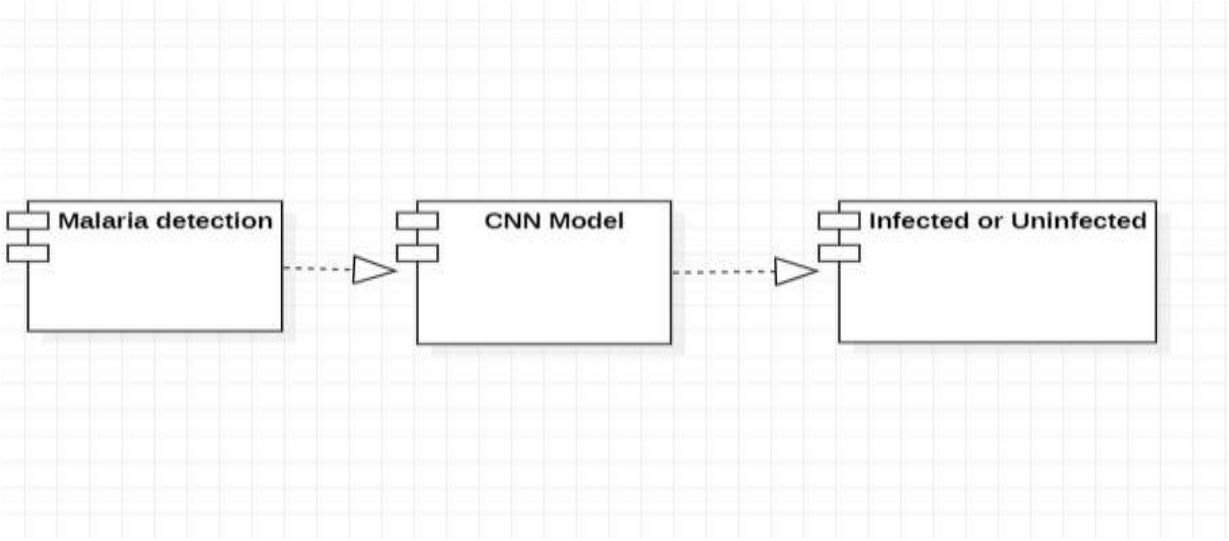



**Fig 3.2.3 Component Diagram**

### 3.2.4 Activity Diagram

An action diagram visually presents a series of actions or flow of control in a system. Activity diagrams are often used in business process modelling.

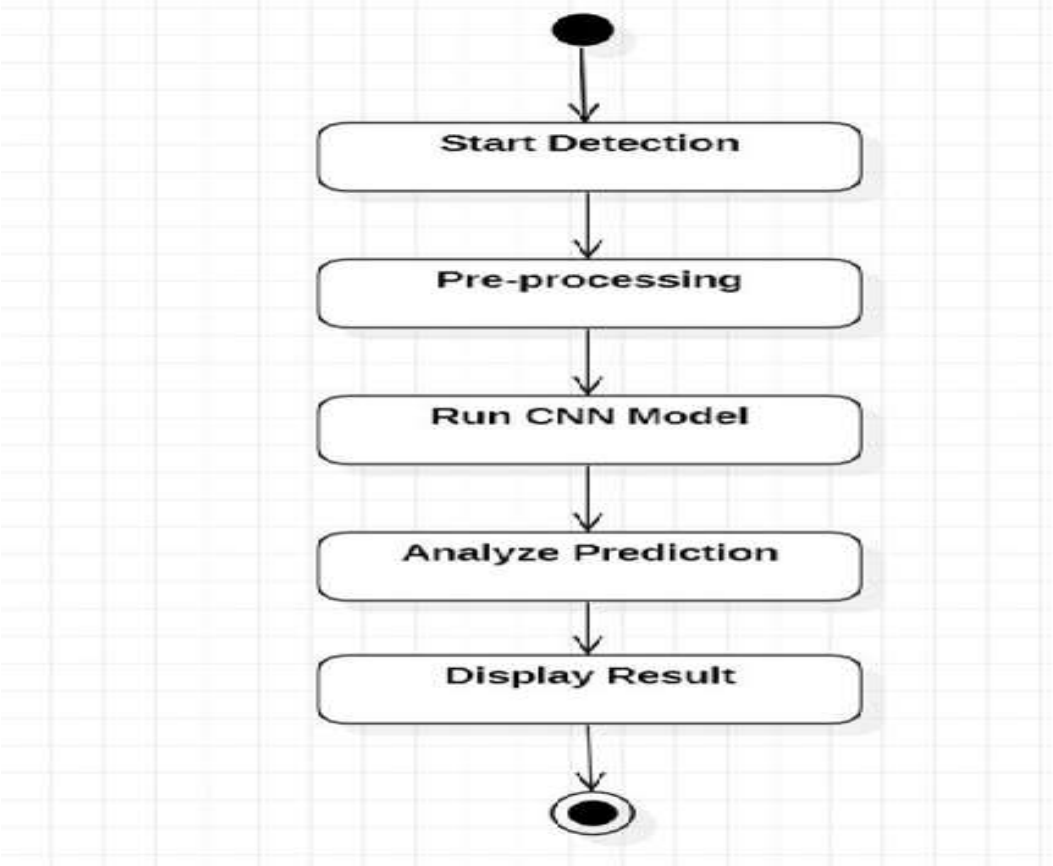



**Fig 3.2.4 Activity diagram**

# CHAPTER –IV

# SYSTEM IMPLEMENTATION

## 4.1 Python

Python is a high-level programming language designed to be easy to read and simple to implement. It is open source, which means it is free to use, even for commercial applications Python is considered a scripting language, like Ruby or Perl and is often used for creating Web applications and dynamic Web content.

Python is dynamically typed and garbage collected. It supports multiple programming paradigms, including structured (particularly, procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.

Guido van Rossum began working on Python in the late 1980s, as a successor to the ABC programming language, and first released it in 1991 as Python 0.9.0. Python 2.0 was released in 2000 and introduced new features, such as list comprehensions and a cycle-detecting garbage collection system (in addition to reference counting). Python 3.0 was released in 2008 and was a major revision ofthe language that is not completely backward-compatible. Python 2 was discontinued with version 2.7.18 in 2020.

Python consistently ranks as one of the most popular programming languages.Python was designed for readability, and has some similarities to the English language with influence from mathematics.Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.

Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curlybrackets for this purpose.It is possible to write Python in an Integrated Development Environment, such as Thonny, Pycharm, Netbeans or Eclipse which are particularly useful when managing larger collections

## Advantages of Python

- **Easy to Read, Learn and Write:** Python is a high-level programming language that has English-like syntax. This makes it easier to read and understand the code.

- **Improved Productivity:** Python is a very productive language. Due to the simplicity of Python, developers can focus on solving the problem.

- **Free and Open-Source:** Python comes under the OSI approved open-source license. This makes it free to use and distribute. You can download the source code, modify and it even distribute version of python. This is useful for organization that want to modify some specific behaviour and use their version for development.

- **Portability:** In many languages like C/C++, you need to change your code to run the program on different platforms. That is not the same with Python. You only write once and run it anywhere.

## 4.2 Sample code

```
from tensorflow.keras.layers import Input, Lambda, Dense, Flatten,Conv2D
from tensorflow.keras.models import Model
from tensorflow.keras.applications.vgg19 import VGG19
from tensorflow.keras.applications.resnet50 import preprocess_input
from tensorflow.keras.preprocessing import image
from tensorflow.keras.preprocessing.image import ImageDataGenerator,load_img
from tensorflow.keras.models import Sequential
import numpy as np
from glob import glob
import matplotlib.pyplot as plt
import PIL
import pathlib
IMAGE_SIZE = [224, 224]
train_path = 'G:\My Drive\Dataset\train'
valid_path = 'G:\My Drive\Dataset\Test'
mobilewnet = VGG19(input_shape=IMAGE_SIZE + [3], weights='imagenet',
include_top=False)
for layer in mobilnet.layers:
    layer.trainable = False
folders = glob(' G:\My Drive\Dataset\train*')
folders
x = Flatten()(mobilenet.output)
prediction = Dense(len(folders), activation='softmax')(x)
model = Model(inputs=mobilnet.input, outputs=prediction)
model.summary()
model=Sequential()
model.add(Conv2D(filters=16,kernel_size=2,padding="same",activation="relu",input_shape
=(224,224,3)))
model.add(MaxPooling2D(pool_size=2))
model.add(Conv2D(filters=32,kernel_size=2,padding="same",activation ="relu"))
model.add(MaxPooling2D(pool_size=2))
```

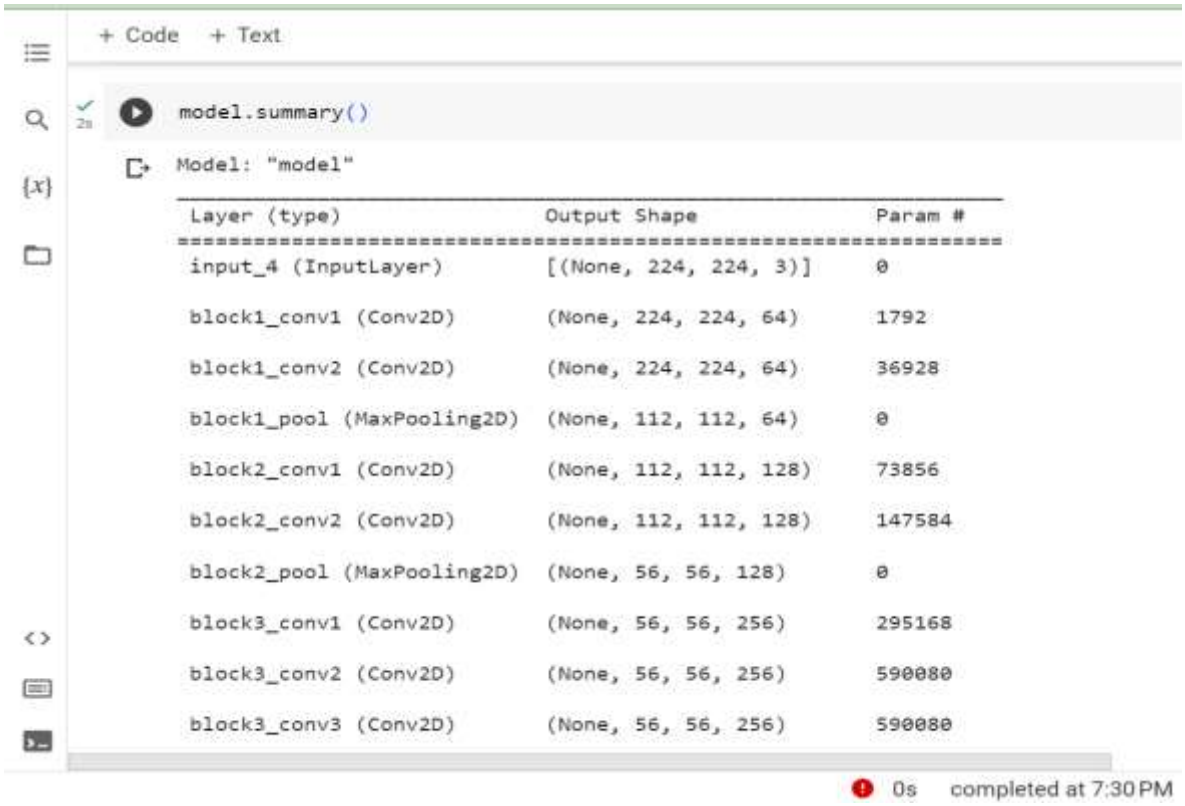
```
model.add(Conv2D(filters=64,kernel_size=2,padding="same",activation="relu"))
model.add(MaxPooling2D(pool_size=2))
model.add(Flatten())
model.add(Dense(500,activation="relu"))
model.add(Dense(2,activation="softmax"))
model.summary()
model.compile(
  loss='categorical_crossentropy',
  optimizer='adam',
  metrics=['accuracy']
)
from tensorflow.keras.preprocessing.image import ImageDataGenerator

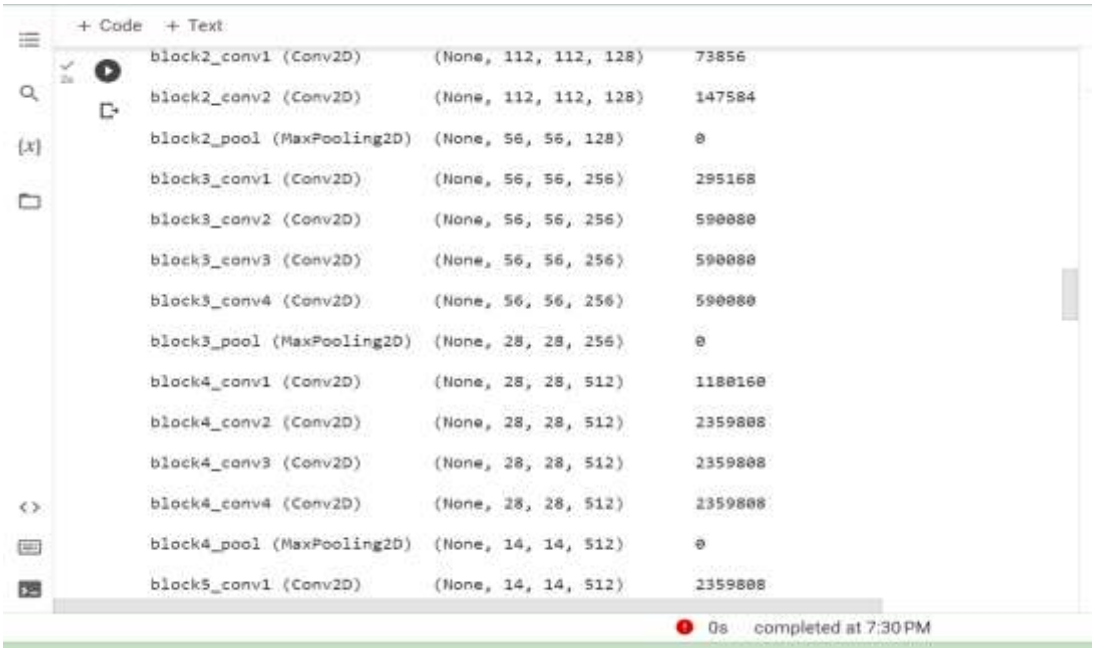
train_datagen = ImageDataGenerator(rescale = 1./255,
                                   shear_range = 0.2,
                                   zoom_range = 0.2,
                                   horizontal_flip = True)
test_datagen = ImageDataGenerator(rescale = 1./255)
training_set  =  train_datagen.flow_from_directory('G:\My Drive\Dataset\train',
                                                   target_size = (224, 224),
                                                   batch_size = 32,
                                                   class_mode = 'categorical')
test_set = test_datagen.flow_from_directory('G:\My Drive\Dataset\Test',
                                            target_size = (224, 224),
                                            batch_size = 32,
                                            class_mode = 'categorical')
r = model..fit_generator(
  training_set,
  validation_data=test_set,
  epochs=50,
  steps_per_epoch=len(training_set),
  validation_steps=len(test_set)
```

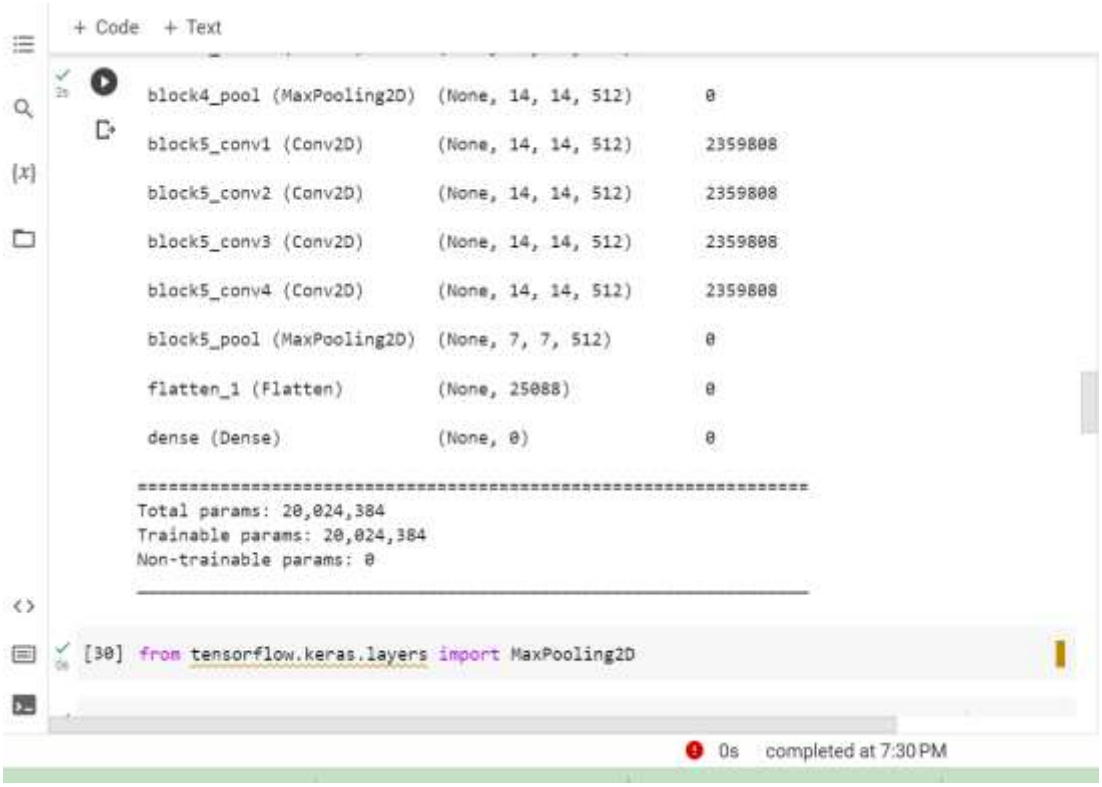
```
)
plt.plot(r.history['loss'], label='train loss')
plt.plot(r.history['val_loss'], label='val loss')
plt.legend()
plt.show()
plt.savefig('LossVal_loss')
plt.plot(r.history['accuracy'], label='train acc')
plt.plot(r.history['val_accuracy'], label='val acc')
plt.legend()
plt.show()
plt.savefig('AccVal_acc')
from tensorflow.keras.models import load_model
model.save('model_vgg19.h5')
y_pred = model.predict(test_set)
y_pred
import numpy as np
y_pred = np.argmax(y_pred, axis=1)
y_pred
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
model=load_model('model_vgg19.h5')
img=image.load_img('Dataset/Test/Uninfected/2.png',target_size=(224,224))
x=image.img_to_array(img)
x
x.shape
x=x/255
x=np.expand_dims(x,axis=0)
img_data=preprocess_input(x)
img_data.shape
model.predict(img_data)
a=np.argmax(model.predict(img_data), axis=1)
if(a==1):
```

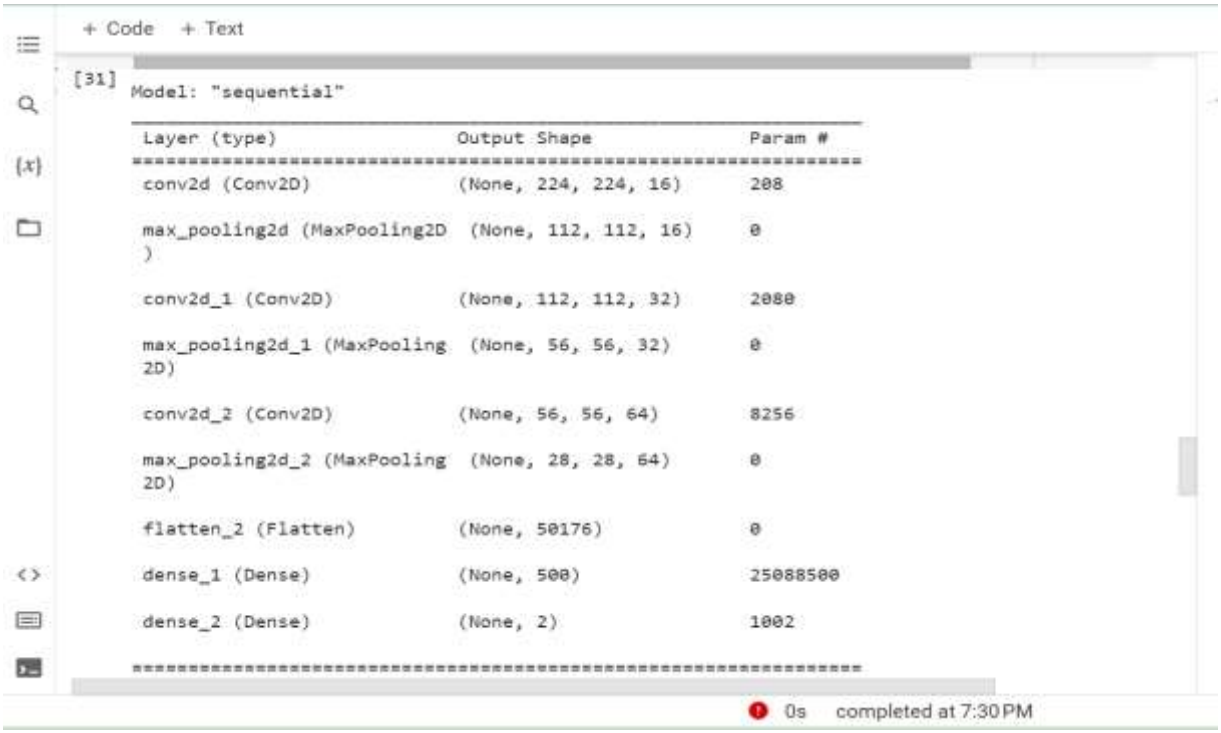
- print("Uninfected")
- **else**:
- print("Infected")

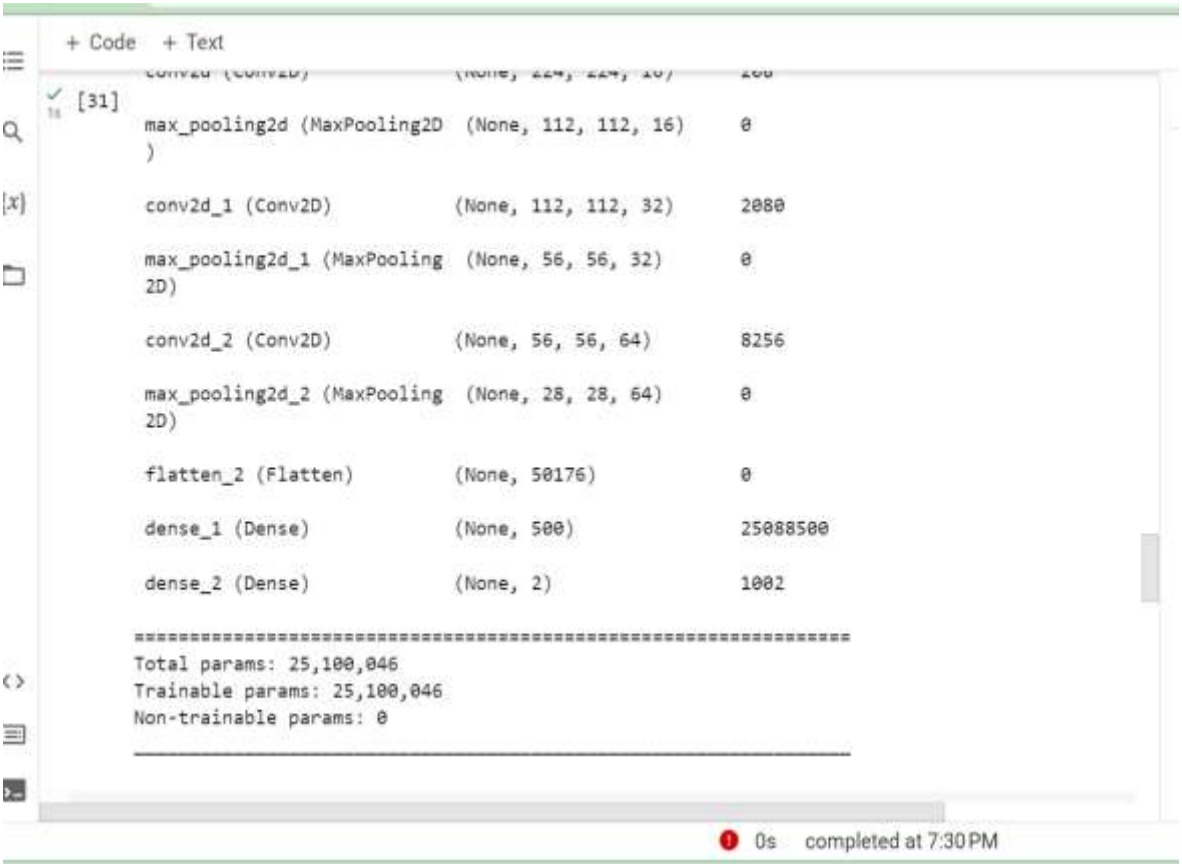
### 4.3 Screenshots

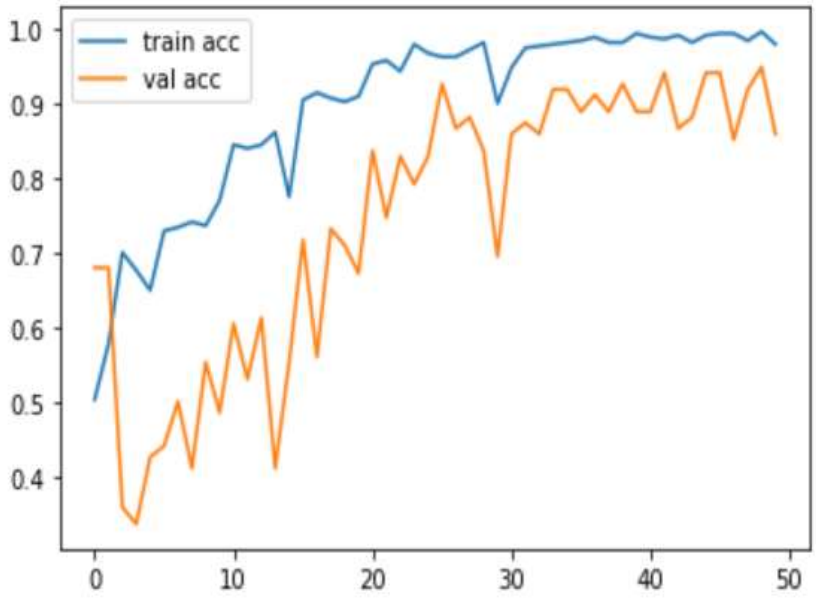



+ Code + Text
block2_conv1 (Conv2D) (None, 112, 112, 128) 73856
block2_conv2 (Conv2D) (None, 112, 112, 128) 147584
block2_pool (MaxPooling2D) (None, 56, 56, 128) 0
block3_conv1 (Conv2D) (None, 56, 56, 256) 295168
block3_conv2 (Conv2D) (None, 56, 56, 256) 590080
block3_conv3 (Conv2D) (None, 56, 56, 256) 590080
block3_conv4 (Conv2D) (None, 56, 56, 256) 590080
block3_pool (MaxPooling2D) (None, 28, 28, 256) 0
block4_conv1 (Conv2D) (None, 28, 28, 512) 1180160
block4_conv2 (Conv2D) (None, 28, 28, 512) 2359808
block4_conv3 (Conv2D) (None, 28, 28, 512) 2359808
block4_conv4 (Conv2D) (None, 28, 28, 512) 2359808
block4_pool (MaxPooling2D) (None, 14, 14, 512) 0
block5_conv1 (Conv2D) (None, 14, 14, 512) 2359808
0s completed at 7:30 PM


+ Code + Text
block4_pool (MaxPooling2D) (None, 14, 14, 512) 0
block5_conv1 (Conv2D) (None, 14, 14, 512) 2359808
block5_conv2 (Conv2D) (None, 14, 14, 512) 2359808
block5_conv3 (Conv2D) (None, 14, 14, 512) 2359808
block5_conv4 (Conv2D) (None, 14, 14, 512) 2359808
block5_pool (MaxPooling2D) (None, 7, 7, 512) 0
flatten_1 (Flatten) (None, 25088) 0
dense (Dense) (None, 0) 0
Total params: 20,024,384
Trainable params: 20,024,384
Non-trainable params: 0
[30] from tensorflow.keras.layers import MaxPooling2D
0s completed at 7:30 PM


+ Code
+ Text
[31]
Model: "sequential"
Layer (type) Output Shape Param #
conv2d (Conv2D) (None, 224, 224, 16) 208
max_pooling2d (MaxPooling2D (None, 112, 112, 16) 0
)
conv2d_1 (Conv2D) (None, 112, 112, 32) 2080
max_pooling2d_1 (MaxPooling (None, 56, 56, 32) 0
2D)
conv2d_2 (Conv2D) (None, 56, 56, 64) 8256
max_pooling2d_2 (MaxPooling (None, 28, 28, 64) 0
2D)
flatten_2 (Flatten) (None, 50176) 0
dense_1 (Dense) (None, 500) 25088500
dense_2 (Dense) (None, 2) 1002
0s completed at 7:30 PM


+ Code
+ Text
[31]
max_pooling2d (MaxPooling2D (None, 112, 112, 16) 0
)
conv2d_1 (Conv2D) (None, 112, 112, 32) 2080
max_pooling2d_1 (MaxPooling (None, 56, 56, 32) 0
2D)
conv2d_2 (Conv2D) (None, 56, 56, 64) 8256
max_pooling2d_2 (MaxPooling (None, 28, 28, 64) 0
2D)
flatten_2 (Flatten) (None, 50176) 0
dense_1 (Dense) (None, 500) 25088500
dense_2 (Dense) (None, 2) 1002
Total params: 25,100,046
Trainable params: 25,100,046
Non-trainable params: 0
0s completed at 7:30 PM




<Figure size 432x288 with 0 Axes>

# CHAPTER -V

# 5. SYSTEM TESTING

**Testing Description**

Testing is the process of evaluating a system or its component(s) with the intent to find whether it satisfies the specified requirements or not. In simple words, testing is executing a system inorder to identify any gaps, errors, or missing requirements in contrary to the actual requirements.

It depends on the process and the associated stakeholders of the project(s). In the IT industry, large companies have a team with responsibilities to evaluate the developed software in context of the given requirements. Moreover, developers also conduct testing which is called Unit Testing. In most cases, the following professionals are involved in testing a system within their respective capacities−software tester, software developer, project lead/manager and end user.

### 5.1.1 Unit Testing

Unit testing focus on the smallest unit of the ware design like module components Test strategy conducted on each module interface to access the flow of input and output. The local data structure is accessible to verify integrity during execution by using unit testing the boundary conditions are tested all error handling.

### 5.1.2 Integration testing

The testing is the systematic technique for constructing the program structure by informing the test in each module and later combining the entire individual module to form a large program.

### 5.1.3 Validation Testing

Validation testing is the process of testing the input. Whether the given inputs are valid or not. When working with databases it is important to validate user's entries, which can be done by using scripting code. The terms verification and validation are used interchangeably we will describe both these methods.

### 5.1.4 Verification Testing

Verification testing is the process of evaluating the work progress and is used to confirm that a process meets the requirements as defined in the phase of the developing the application process.

# CHAPTER –VI

# CONCLUSION & FUTURE ENHANCEMENTS

## 6.1 Conclusion:

Malaria causes large number of facilities every year and poses a particular threat to younger people. The CNN deep learning approach offers a means of producing effective image classification models which might be well suited to medical applications, such as malaria detection and diagnosis. The classification accuracy of CNN can be improved by the application of an activation function. The effectiveness of the CNN model could be significantly enhanced depending on the number of images involved and the choice of data preprocessing methods.

## 6.2 Future Enhancements:

Future enhancement for malaria detection using computational technique significantly contribute to improving accuracy,efficiency and accessibility of diagnostic methods.Some potential future enhancements are deep learning models,Transfer learning, Data augmentation,Multi-modal integration,Automated parasite counting,Telemedicine integration. By pursuaing these future enhancements,malaria detection using computational methods can achieve even higher accuracy, accessibikity and scalability. So it can contribute to the timely identification andtreatment of malaria cause,playing role in global efforts to combat this deadly disease.

.

# REFERENCES


1. Saeed,S.,Saqib,M.,&Jan,Z,(2019),Malaria detection using deep learning.2019 IEEE 5th International Conference on Engineering Technologies and Applied Science(ICETAS).
2. Haque,M,N.,Hasan,M,M.,&Mahmud,M(2020),Automatic malaria detections using convolutional neural networks,2020 International Conference on Computer,Communication,Chemical,Materials and Electronic Engineering(IC4ME2).
3. Akram,M,U.,Ali,T,Saba,T.,Umar,M.,& Nabi, G,(2020). A review on deep learning techniques for malaria detection from microscopicblood smear images. Computers in Biology and Medicine.
4. Mohammed HA, Abdelrahman IAM. International conference on communication, control, computing and electronics engineering *(ICCCCEE)* IEEE; 2017. Detection and classification of malaria in thin blood slide images
5. A. Simon, R. Vinayakumar, V. Sowmya, and K. P. Soman, "Shallow cnn with lstm layer for tuberculosis detection in microscopic images," Machine learning for Biomedical Applications, 2019.
6. D. Yang, G. Subramanian, J. Duan et al., "A portable image-based cytometer for rapid malaria detection and quantification," PLoS One, vol. 12, no. 6, Article ID e0179161, 2017.
7. H. A. Mohammed and I. A. M. Abdelrahman, "Detection and classification of malaria in thin blood slide images," in Proceedings of the 2017 International Conference on Communication, Control, Computing and Electronics Engineering (ICCCCEE), pp. 1–5, Khartoum, Sudan, January 2017.
8. S. M. Parsel, S. A Gustafson, E Friedlander et al., "Malaria over-diagnosis in Cameroon: diagnostic accuracy of Fluorescence and Staining Technologies (FAST) Malaria Stain and LED microscopy versus Giemsa and bright field microscopy validated by polymerase chain reaction," Infectious diseases of poverty, vol. 6, no. 1, pp. 32–39, 2017.
9. C. Wongsrichanalai, F. Kawamoto, M. Hommel, and P. G. Kremsner, "Fluorescent microscopy and fluorescent labelling for malaria diagnosis," *Encycl. Malar*, Springer, New York, NY, USA, pp. 1–7, 2021.
10. K. M. F. Fuhad M. Umer, S. Sadiq, M. Ahmad, S. Ullah, G. S. Choi, and A. Mehmood, "A novel stacked cnn for malarial parasite detection in thin blood smear images," *IEEE Access*, vol. 8, pp. 93782–93792, 2020.

11. A. Vijayalakshmi and B. R. Kanna, “Deep learning approach to detect malaria from microscopic images,” Multimedia Tools and Applications, vol. 79, no. 21, pp. 15297–15317, 2020.

12. Rajaraman, Antani,S.,&Poostchi,M,(2019),Malaria parasite detection in thin blood smear images using deep convolutional neural networks . In Proceedings of the IEEE conference on computer vision and pattern recognition workshops.

# PUBLICATION